



Medication management, health outcomes, and economic burden of severe asthma stratified by asthma control levels

Ke Zhang^{a,b}, Xiaoning He^{a,b}, Jing Wu^{a,b,*}

^a School of Pharmaceutical Science and Technology, Faculty of Medicine, Tianjin University, Tianjin, China

^b Center for Social Science Survey and Data, Tianjin University, Tianjin, China

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ABSTRACT

Background: Severe asthma—a critical subset of asthma—is known for escalating the risk of asthma-related symptoms. However, current research on the management and burden of severe asthma in developing countries remains limited, especially in China, constraining evidence-based policymaking.

Objectives: This study aims to evaluate the medication management, health outcomes, and economic burden of severe asthma across different control levels in China.

Methods: We analyzed routinely collected data on nearly 1.7 million subjects from Tianjin, China. Severe asthma was identified based on specific medication use patterns. Patients were stratified as uncontrolled, suboptimally controlled or controlled according to the inpatient admission, short-term systemic corticosteroid use and short-acting inhaled beta₂-agonist use. Over a one-year follow-up period, the medication management, health outcomes, and economic burden were assessed.

Results: A total of 2418 patients with severe asthma were identified. Inhaled corticosteroids/long-acting beta₂-agonist (ICS/LABA) was the most commonly used controller medication (93.51 %), while patients with poorer control spent more on additional controller medications and Chinese herbal medicines. Oral corticosteroids (OCS) were less used (10.75 %), but the mean daily dosage of OCS among OCS users was as high as 15.68 mg—1.6 times the recommended dosage in Chinese guidelines. Only 17.78 % of patients with severe asthma demonstrated good adherence (defined as proportion of days covered ≥ 0.8) to asthma control treatment. Patients with uncontrolled asthma showed a significantly high proportion of exacerbations (51.42 %) and incurred higher asthma-related medical cost than suboptimally controlled and controlled patients (CNY 8506.83 vs. 3559.35 vs. 2105.70, $p = 0.000$).

Conclusion: Patients with severe asthma exhibit poor treatment adherence, low OCS usage rates but excessively high doses among users, pursuit of alternative therapies, and a high economic burden in Tianjin, China. These findings underscore the urgent need to improve severe asthma management, particularly for those with uncontrolled asthma.

Introduction

Asthma is a chronic airway inflammatory disease clinically characterized by symptoms such as shortness of breath, wheezing, coughing, chest tightness, and variable airflow limitation. The World Health Organization reported that asthma affected 262 million people globally in 2019, underscoring its widespread impact.¹ The global prevalence of asthma is rising, driven in part by population aging.² Severe asthma is a critical subset of asthma, accounting for approximately 5 % to 10 % of asthma patients globally. It contributes 41 % to 51 % of the total

economic burden of asthma.^{3–5} Definitions of severe asthma vary between countries; in China, it is defined as a condition requiring medium-to-high dose inhaled corticosteroid (ICS) combined with additional controllers, such as long-acting beta₂-agonist (LABA), to achieve control.⁶ Poor asthma control is associated with an elevated risk of exacerbations, increased emergency department visit, hospitalization, and disability.^{7–9}

Despite its growing prevalence, the medication management, health outcomes, and economic burden of severe asthma remain underexplored, particularly given the variability in clinical characteristics, treatment practices, economic environments across countries.^{10,11}

* Corresponding author at: Professor; School of Pharmaceutical Science and Technology, Faculty of Medicine, Tianjin University; Center for Social Science Survey and Data, Tianjin University, Address: No. 92 Weijin Road, Nankai District, Tianjin University, Tianjin 300072, China.

E-mail address: jingwu@tju.edu.cn (J. Wu).

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Abbreviations	
ICS	Inhaled corticosteroid
LABA	Long-acting beta ₂ -agonist
OCS	Oral corticosteroids
HIS	Hospital information systems
GINA	Global Initiative for Asthma
LTRA	Leukotriene receptor antagonist
LAMA	Long-acting muscarinic antagonist
SCS	Systemic corticosteroid
SABA	Short-acting inhaled beta ₂ -agonist
CCI	Charlson Comorbidity Index
SAMA	Short-acting muscarinic antagonist
PDC	Proportion of days covered

Existing research has predominantly focused on developed regions, including the US, the UK, Italy, and Japan, with limited investigations in developing countries such as China. Moreover, few studies have examined the economic burden of severe asthma in relation to asthma control, and those that have typically involve small sample sizes.^{12,13} Current studies also tend to emphasize prescribed medications for severe asthma, with insufficient attention to treatment adherence or oral corticosteroids (OCS) use, both critical for effective management. The knowledge gaps regarding medication management, health outcomes, and economic burden of severe asthma across different control levels in China hinder evidence-based policymaking.

This study aims to assess medication management, health outcomes, and economic burden—including direct healthcare utilization and associated cost—among patients with severe asthma based on regional medical data from Tianjin, China. Tianjin is a provincial-level city with a population exceeding 10 million. It has moderate levels of healthcare resources and health expenditure compared to other regions in China. For example, the number of health technical personnel per 1000 persons is 8.87, which is close to the national average of 7.97.¹⁴ Similarly, the per capita health expenditure is CNY 6545, slightly higher than the national average of CNY 5111.¹⁴ These characteristics may enhance the generalizability of the findings of Tianjin to other regions of China. Findings from this study could also provide valuable insights for other developing countries with similar healthcare settings.

Methods

Study population and data collection

This cohort study with a 1-year follow-up used the Tianjin regional medical database, comprising data routinely collected prior to study initiation. This database includes nearly 1.7 million randomly sampled subjects from the hospital information systems (HIS) of over 90 tier-three and tier-two hospitals, which are the primary healthcare providers for residents in Tianjin, China. The database covers the period from 2018 to 2020 and gathers information on outpatient, emergency department and inpatient visits of patients. Core variables in the database include sociodemographic information (e.g., age and sex), visit information (e.g., dates of visit and discharge, diagnostic text and codes [based on the 10th revision of the International Classification of Diseases, ICD-10], visit type), treatment data (e.g., services, medication names, specifications, usage, dosage), and cost information (e.g., total visit costs, cost breakdowns). Patient records across different institutions can be linked using a unique de-identified code. This study was approved by the Ethics Committee of Tianjin University (Approval number: TJUE-2024-011).

Research period and patients

The identification period was from January 1, 2019 to December 31, 2019. Patients with at least one asthma diagnosis, identified through ICD-10 code J45, J46 or diagnostic text, were initially included. The index date was defined as the date of the first asthma diagnosis during the identification period. The 12 months prior to the index date constituted the baseline period, and 12 months following the index date comprised the follow-up period.

Severe asthma was defined in accordance with the Guidelines for Bronchial Asthma Prevent and Management (2020 edition) in China,⁶ which aligns with Global Initiative for Asthma (GINA) steps 4 or 5—requiring such treatment intensity to achieve asthma control.¹⁵ Patients were identified as having severe asthma if, during the baseline period, they had been prescribed at least two courses of medium-to-high dose ICS combined with at least one additional controller (prescribed within 30 days before or after the ICS prescription), with prescriptions linked to an asthma diagnosis. Additional controllers included LABA, leukotriene receptor antagonist (LTRA), long-acting muscarinic antagonist (LAMA) and theophylline. Patients with incomplete follow-up records (excluding deaths) were excluded. Eligible patients with severe asthma were then followed for 12 months.

Given the lack of a standardized definition of asthma control based on healthcare resource utilization in China, patients with severe asthma were categorized into "controlled," "suboptimally controlled," or "uncontrolled" asthma using criteria consistent with the American Thoracic Society/European Respiratory Society guidelines and published studies.^{16–18} During the follow-up period, patients were classified as follows: (i) Uncontrolled asthma: at least one inpatient admission with a primary asthma diagnosis, at least two instances of short-term systemic corticosteroid (SCS) use, or at least four prescriptions for short-acting inhaled beta₂-agonist (SABA); (ii) Suboptimally controlled asthma: at least one instance of short-term SCS use or one to three prescriptions for SABA, (iii) Controlled: patients who did not meet the above criteria. All medications were required to be prescribed in visits with an asthma diagnosis. Short-term SCS use was defined as OCS use from three days to one week, or at least one administration of injectable corticosteroids.

Outcome measures

Patient characteristics included age, sex, Charlson Comorbidity Index (CCI) and asthma-related comorbidities. Asthma medications used during the entire follow-up period (including exacerbation periods) and during only exacerbation periods were recorded separately. These included controllers for asthma control (e.g., ICS/LABA, LTRA) and relievers for symptom relief (e.g. SABA, short-acting muscarinic antagonist [SAMA]). The prescription of Chinese herbal medicines linked to asthma diagnoses was also assessed, including Chinese medicinal materials (medicinal parts of medicinal plants, animals, and minerals after preliminary processing), decoction pieces (prescription medicinal processed from Chinese medicinal materials) and Chinese patent medicines (standardized preparation formulated based on specific prescriptions and preparation processes).^{19,20} Adherence to asthma control treatment, which included ICS in combination with LABA, LTRA, LAMA, and theophylline, was assessed using the proportion of days covered (PDC). Good adherence was defined as PDC ≥ 0.8. Additionally, the duration and dosage of OCS use (in prednisone equivalents) were also calculated.

Health outcomes included the proportion of patients with exacerbation, number of exacerbations per patient, and mortality. Exacerbations were defined as inpatient admissions with a primary asthma diagnosis, emergency department visits with an asthma diagnosis, or short-term SCS use with an asthma diagnosis.⁶ Multiple exacerbations occurring within one week were combined as a single exacerbation and classified by severity (inpatient admission > emergency department visit > short-term use of SCS).

All-cause and asthma-related healthcare resource utilization and

costs during the entire follow-up period were estimated from the healthcare system perspective. Asthma-related resource utilization and costs included those associated with outpatient and emergency department visits with an asthma diagnosis, and inpatient admissions with a primary asthma diagnosis. Costs of different medication types were also calculated. Additionally, healthcare resource utilization and costs due to exacerbations were analyzed.

Statistical analysis

All outcomes were stratified by asthma control levels. Daily dosage of ICS was calculated by multiplying the number of administrations per day by the dosage per administration. Missing values in the number of administrations and the dosage were imputed using hierarchical imputation based on observed data, ordered by similarity in individual characteristics, medication, and specifications. Data are presented as mean $\pm$ standard deviation or as absolute values (percentages). Continuous variables were compared using ANOVA or the Kruskal-Wallis test; categorical variables were compared using Pearson's Chi-square test or Fisher's exact test, as appropriate. A p-value < 0.05 was considered statistically significant. All analyses were performed using Stata 17.0 and SPSS 27.0.

Patient and public involvement

Patients and the public were not involved in this analysis of previously collected data.

Results

Patient characteristics

A total of 32,084 subjects were diagnosed with asthma, with 8.99 % excluded due to incomplete follow-up records. Among the remaining, 2418 patients (7.54 %) were identified as having severe asthma (Fig. 1). Of these, 212 (8.77 %) were classified as having uncontrolled asthma, 298 (12.32 %) as suboptimally controlled, and 1908 (78.91 %) as controlled.

The mean age of patients with severe asthma at the index date was 56.15 ± 14.39 years, and 44.38 % were female. Poorer asthma control was significantly associated with older age (uncontrolled vs. suboptimally controlled vs. controlled: 58.30 ± 12.54 vs. 57.47 ± 14.00 vs. 55.70 ± 14.61 , $p = 0.026$), had higher CCI scores (3.13 ± 2.37 vs. 2.69 ± 2.13 vs. 2.46 ± 2.06 , $p < 0.001$), and a higher prevalence of comorbidities including COPD (42.45 % vs. 38.93 % vs. 32.08 %, $p = 0.001$),

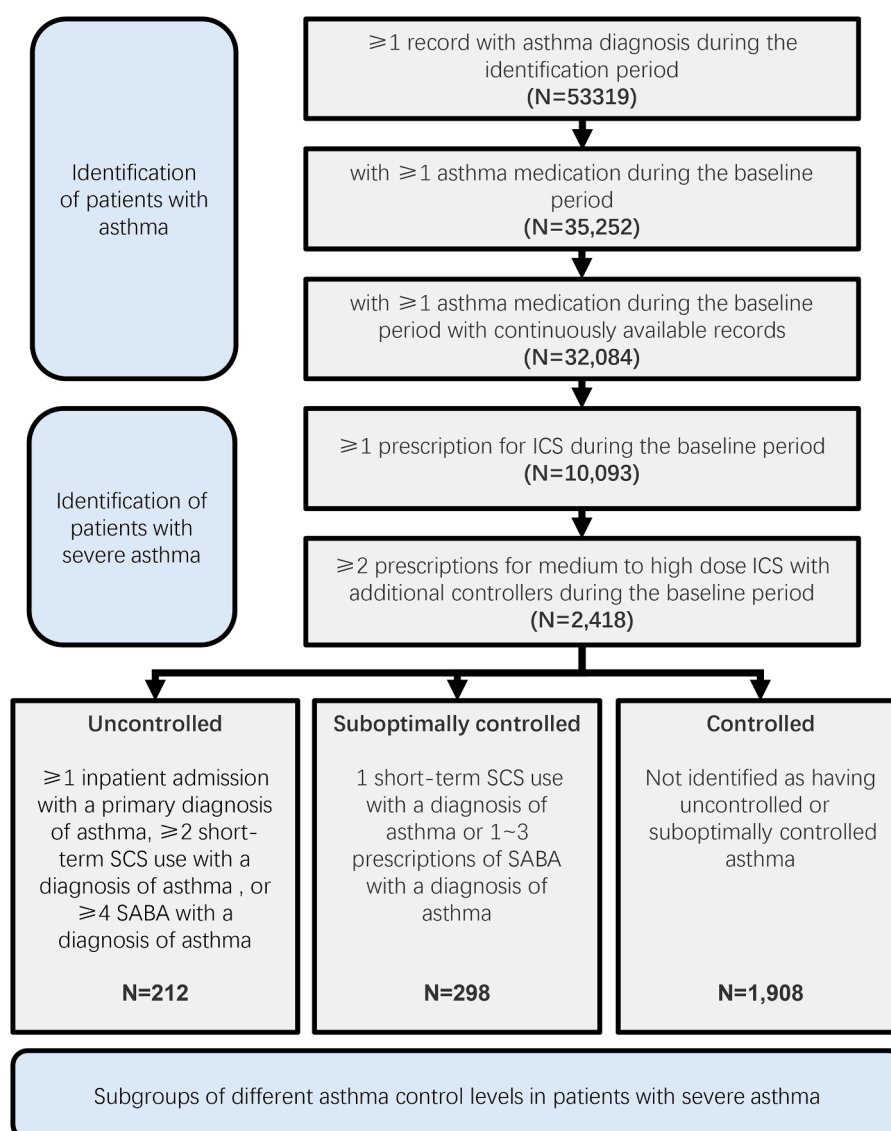


Fig. 1. Patient selection.

allergic rhinitis (31.60 % vs. 38.26 % vs. 30.92 %, $p = 0.042$), and anxiety (17.45 % vs. 19.13 % vs. 12.63 %, $p = 0.003$). The baseline characteristics of patients with severe asthma are shown in Table 1. Patients with uncontrolled or suboptimally controlled asthma demonstrated higher healthcare resource utilization and costs during the baseline period. Additional details are provided in eTable 1.

Medication management

As shown in Fig. 2, ICS/LABA was the most commonly used controller throughout the entire follow-up period (93.51 %), followed by LTRA (56.16 %) and cough medicine (49.05 %). Only 2.11 % of patients used OCS ≥ 180 days. SABA was the most frequent reliever (23.61 %). Compared to controlled patients, those with uncontrolled asthma had significantly higher usage rates for most medications, except ICS/LABA, anti-allergics, and LAMA. Patients with suboptimally controlled asthma demonstrated moderately higher usage proportions. During the exacerbation period (detailed in eTable 2), both uncontrolled and suboptimally controlled patients had significantly higher reliever usage than controlled patients.

A total of 50.79 % of patients were prescribed at least one Chinese herbal medicines for asthma; among these patients, Chinese patent medicines were the most commonly used type (96.58 %). Among patients with uncontrolled asthma, both SuXiao JiuXin Wan (Quick-Acting Help Heart Pills) and Fei Li cough mixture were the most commonly prescribed, each used by 25 % of patients. Additional details are provided eTable 3.

During the entire follow-up period, the mean adherence to asthma control treatment among all patients was 0.44 ± 0.36 , with only 17.78 % of them demonstrating good adherence ($PDC \geq 0.8$). Adherence increased with poorer control (uncontrolled: 0.54 ± 0.34 vs. suboptimally controlled: 0.48 ± 0.32 vs. controlled: 0.42 ± 0.37 , $p < 0.001$). The distribution of adherence ranges is shown in eTable 4.

A total of 260 (10.75 %) patients used OCS during the entire follow-up period. Among these patients, the mean duration of OCS use was 82.73 ± 108.64 days, with 19.62 % using OCS for >180 days. On average, patients taking OCS had a daily dosage of 15.68 ± 6.17 mg, with a mean total dosage of 1424.20 ± 2051.69 mg. Patients with uncontrolled or suboptimally controlled asthma had significantly longer durations of OCS use compared to patients with controlled asthma (uncontrolled: 111.21 days, suboptimally controlled: 102.27 days, controlled: 53.35 days, $p < 0.001$), resulting in substantially higher total dosages over the follow-period. Detailed information is provided in eTable 5.

Health outcomes

A total of 258 (10.67 %) patients experienced at least one exacerbation during the entire follow-up period, with a mean number of

exacerbations of 0.21 per patient. Specifically, 163 patients (6.74 %) had one exacerbation, and 95 (3.93 %) had two or more exacerbations. Patients with uncontrolled asthma had a significantly higher exacerbation proportion (51.42 %) and a greater mean number of exacerbations (1.31) compared to the other two groups. They also had a higher proportion of patients with two or more exacerbations (26.89 %). Further details are available in eTable 6.

The overall mortality was 1.16 %, with a significant difference observed across different asthma control levels (uncontrolled: 1.42 % vs. suboptimally controlled: 2.68 % vs. controlled: 0.89 %, $p = 0.025$).

Healthcare resource utilization and costs

Nearly all patients had at least one all-cause outpatient visit (99.96 %), while 15.01 % of all patients had inpatient admissions. The mean all-cause direct medical cost per patient during the follow-up period was CNY $18,031.93 \pm 32,821.24$ (detailed in eTable 7). Asthma-related outpatient visits were recorded for 99.79 % of patients, with an average of 8.18 ± 8.67 visits. Patients with worse asthma control had significantly higher numbers of both outpatient and emergency department visits. Inpatient admissions were observed in 1.28 % of patients, all of whom had uncontrolled asthma. The mean asthma-related medical cost per patient was CNY 2846.08 ± 4677.92 , with 87.52 % of the total cost attributed to outpatient medications. Patients with uncontrolled asthma showed higher asthma-related medical cost compared to the two other groups (uncontrolled: CNY $8506.83 \pm 11,264.60$ vs. suboptimally controlled: CNY 3559.35 ± 4313.70 vs. controlled: CNY 2105.70 ± 2570.42 , $p = 0.000$). Detailed results are shown in Table 2.

As shown in Fig. 3, outpatient medication costs were distributed as follows: controllers accounted for 59.25 % (CNY 1475.85), relievers for 1.28 % (CNY 31.89), and other medications for 39.47 % (CNY 983.06). ICS/LABA made up 70.90 % of the outpatient medication costs. As asthma control level worsened, the proportions of controller costs decreased (uncontrolled: 40.40 % vs. suboptimally controlled: 51.61 % vs. controlled: 67.31 %, $p < 0.001$), while spending on relievers (4.46 % vs. 1.85 % vs. 0.11 %, $p = 0.000$) and other medications (55.14 % vs. 46.55 % vs. 32.58 %, $p = 0.000$) increased. Meanwhile, the proportions of ICS/LABA among controllers also decreased along with worsening control (61.15 % vs. 65.37 % vs. 73.89 %, $p = 0.006$). Chinese herbal medicine accounted for 16.65 % (CNY 414.75) of the outpatient medication costs. Patients with uncontrolled asthma spent a larger proportion on Chinese herbal medicine (24.58 %) compared to those with suboptimally controlled asthma (20.42 %) or controlled asthma (13.12 %).

Among patients with exacerbations, those experienced exacerbations requiring the emergency department visit were the most common (67.05 %), followed by patients requiring the short-term use of SCS (26.74 %) and those requiring the inpatient admission (12.02 %). Patients with

Table 1
Patient characteristics.

	Total (N = 2418)	Controlled (N = 1908)	Suboptimally Controlled (N = 298)	Uncontrolled (N = 212)	P-value
Age (years)	56.15 $\pm$ 14.39	55.70 $\pm$ 14.61	57.47 $\pm$ 14.00	58.30 $\pm$ 12.54	0.026
Sex					
male	1345 (55.62 %)	1072 (56.18 %)	156 (52.35 %)	117 (55.19 %)	0.462
female	1073 (44.38 %)	836 (43.82 %)	142 (47.65 %)	95 (44.81 %)	
CCI	2.55 $\pm$ 2.10	2.46 $\pm$ 2.06	2.69 $\pm$ 2.13	3.13 $\pm$ 2.37	<0.001
Comorbidities					
COPD	818 (33.83 %)	612 (32.08 %)	116 (38.93 %)	90 (42.45 %)	0.001
Allergic rhinitis	771 (31.89 %)	590 (30.92 %)	114 (38.26 %)	67 (31.60 %)	0.042
Osteoporosis	344 (14.23 %)	272 (14.26 %)	34 (11.41 %)	38 (17.92 %)	0.116
Anxiety	335 (13.85 %)	241 (12.63 %)	57 (19.13 %)	37 (17.45 %)	0.003
Gastrointestinal ulcers/bleeds	308 (12.74 %)	244 (12.79 %)	42 (14.09 %)	22 (10.38 %)	0.461
Depression	248 (10.26 %)	190 (9.96 %)	35 (11.74 %)	23 (10.85 %)	0.61

Definition of abbreviations: CCI= Charlson Comorbidity Index, COPD=chronic obstructive pulmonary disease.

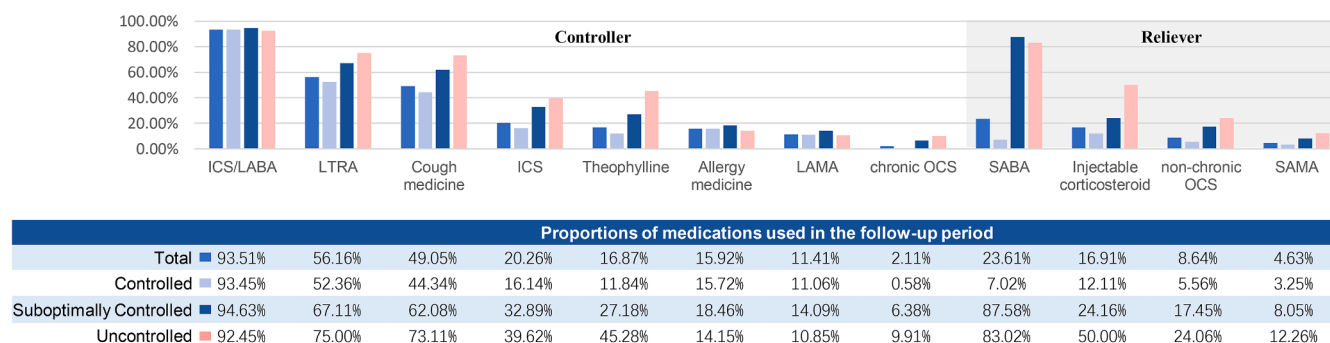


Fig. 2. Proportions of medications used in the follow-up period. *All OCS use in the exacerbation period was classified as reliever due to the definition of exacerbation. Definition of abbreviations: ICS/LABA=inhaled corticosteroids/long-acting beta₂-agonist, LTRA=leukotriene receptor antagonist, ICS= inhaled corticosteroids, LAMA=long-acting muscarinic antagonist, OCS=oral corticosteroid, SABA= short-acting beta₂-agonist, SAMA=short-acting muscarinic antagonist.

Table 2

Asthma-related healthcare utilization and costs of the follow-up period.

	Total (N = 2418)	Controlled (N = 1908)	Suboptimally Controlled (N = 298)	Uncontrolled (N = 212)	P-value
Healthcare resource utilization					
Outpatient visit					
Patient with an outpatient visit ^a	2413 (99.79 %)	1904 (99.79 %)	298 (100.00 %)	211 (99.53 %)	0.654
Patient with an emergency department visit	179 (7.40 %)	78 (4.09 %)	37 (12.42 %)	64 (30.19 %)	<0.001
Number of outpatient visits ^a	8.18 ± 8.67	6.73 ± 1.13	10.40 ± 10.01	18.02 ± 15.88	0.000
Number of emergency department visits	0.20 ± 0.99	0.09 ± 0.61	0.15 ± 0.51	1.19 ± 2.52	0.000
Inpatient admission					
Patient with an inpatient admission	31 (1.28 %)	0 (0.00 %)	0 (0.00 %)	31 (14.62 %)	<0.001
Patient with an ICU admission	0 (0.00 %)	0 (0.00 %)	0 (0.00 %)	0 (0.00 %)	1.000
Number of inpatient admissions	0.02 ± 0.29	0.00 ± 0.00	0.00 ± 0.00	0.25 ± 0.94	0.000
Length of stay per admission (days)	0.21 ± 2.45	0.00 ± 0.00	0.00 ± 0.00	2.40 ± 7.97	0.000
Costs (CNY)					
Total direct medical cost per patient	2846.08 ± 4677.92	2105.70 ± 2570.42	3559.35 ± 4313.70	8506.83 ± 11,264.60	0.000
Direct medical outpatient cost per patient	2632.86 ± 3643.90	2105.70 ± 2570.42	3559.35 ± 4313.70	6074.91 ± 7110.81	0.000
Medication	2490.81 ± 3573.46	1993.99 ± 2474.70	3345.20 ± 4251.05	5761.16 ± 7174.95	0.000
Examination	107.06 ± 312.67	90.38 ± 282.42	167.68 ± 412.84	171.95 ± 383.60	<0.001
Service	18.03 ± 115.11	10.74 ± 100.25	25.32 ± 130.50	73.42 ± 182.60	0.000
Medical material	13.62 ± 78.95	9.55 ± 73.75	15.21 ± 56.52	48.01 ± 128.08	0.000
Surgery	0.72 ± 24.69	0.48 ± 20.61	2.73 ± 47.21	0.07 ± 1.03	0.342
Bed	1.91 ± 18.61	0.19 ± 3.33	2.44 ± 10.40	16.63 ± 58.91	0.000
Other	0.71 ± 7.76	0.38 ± 7.73	0.77 ± 3.95	3.68 ± 10.89	0.000
Direct medical inpatient cost per patient	213.22 ± 2885.62	0.00 ± 0.00	0.00 ± 0.00	2431.92 ± 9484.82	0.000
Medication	88.23 ± 1387.14	0.00 ± 0.00	0.00 ± 0.00	1006.37 ± 4594.86	0.000
Examination	59.74 ± 766.17	0.00 ± 0.00	0.00 ± 0.00	681.37 ± 2509.71	0.000
Service	36.29 ± 501.67	0.00 ± 0.00	0.00 ± 0.00	413.96 ± 1650.99	0.000
Medical material	15.88 ± 236.49	0.00 ± 0.00	0.00 ± 0.00	181.14 ± 781.37	0.000
Surgery	0.79 ± 17.15	0.00 ± 0.00	0.00 ± 0.00	8.99 ± 57.40	<0.001
Bed	11.19 ± 148.77	0.00 ± 0.00	0.00 ± 0.00	127.58 ± 488.48	0.000
Other	1.10 ± 14.98	0.00 ± 0.00	0.00 ± 0.00	12.51 ± 49.27	0.000

^a includes emergency department visit.

uncontrolled or suboptimally controlled asthma had higher costs per patient and per exacerbation. Further details are available in eTable 8.

Discussion

Our study estimated the prevalence of severe asthma among patients with asthma in Tianjin to be 7.54 %, with 8.77 % of these patients classified as having uncontrolled asthma. ICS/LABA was the most common medication for asthma. However, adherence to asthma control treatments requires significant improvement, as only a limited number of patients demonstrated good adherence. Patients with uncontrolled or suboptimally controlled asthma exhibited significantly poorer health outcomes and higher economic burden, emphasizing the urgent need to improve asthma control levels.

The estimated prevalence of severe asthma in Tianjin aligns with previous studies in China (3.4 %–8.4 %) and other global reports.^{3,21–23} The distribution of patients across different asthma control levels was

generally consistent with findings from previous studies in the US and Japan (uncontrolled: 5.03 %–5.33 %, suboptimally controlled: 4.70 %–15.52 %),^{17,13} which also assessed asthma control levels based on healthcare resource utilization.

ICS/LABA was the most commonly used control treatment in Tianjin. However, the proportion of ICS/LABA costs declined with worsening asthma control, leading to increased expenditures on alternative control treatments. The pursuit of alternative treatments partially reflects a demand for more effective therapies. On the other hand, theophylline, another low-cost asthma control medication recommended in Chinese guidelines but not recommended in GINA mainly due to metabolic differences across races,⁶ was used by 16.87 % of patients with severe asthma in Tianjin, China. This proportion is significantly higher than those reported in France and the US (1.5 %–6.4 %),^{17,24} highlighting the influence of racial and economic factors.

A total of 10.75 % of patients in Tianjin, China used OCS, and 2.11 % used chronic OCS, which is similar to that observed in Hong Kong, China

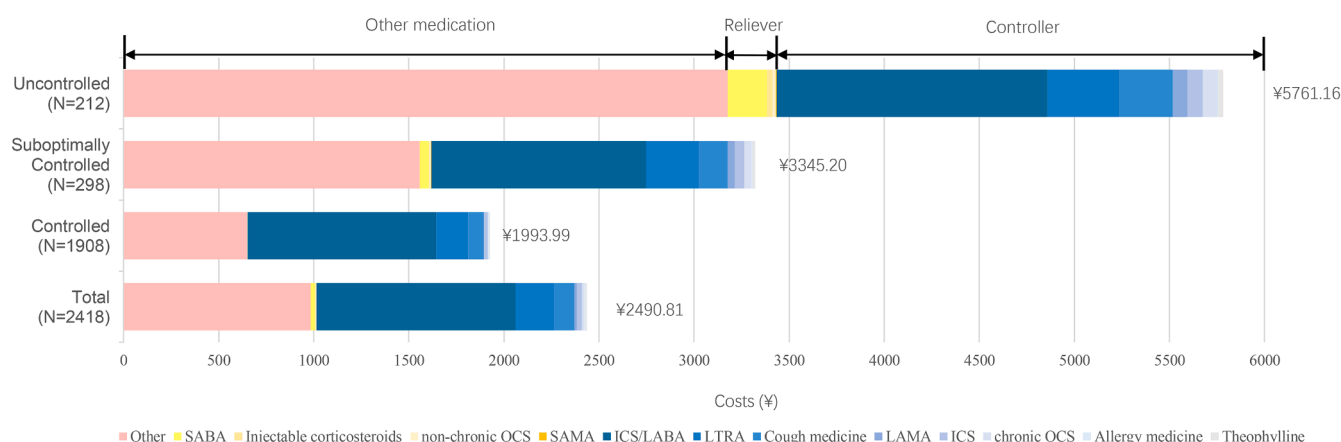


Fig. 3. Costs for each type of medication among the outpatient medication costs. Definition of abbreviations: ICS/LABA=inhaled corticosteroids/long-acting beta₂-agonist, LTRA=leukotriene receptor antagonist, ICS= inhaled corticosteroids, LAMA=long-acting muscarinic antagonist, OCS=oral corticosteroid, SABA= short-acting beta₂-agonist, SAMA=short-acting muscarinic antagonist.

(3.4 %).²⁵ The proportion of OCS use in China was lower than that reported in other countries and regions mainly including Europe, Australia, Canada and the US and Japan (OCS:14 %~48.76 %, chronic OCS: 9.96 %~60.83 %).^{17,24,26–29} One potential factor may be varying levels of corticosteroid-related concerns or "phobia" across regions, as is partly evident by that Chinese patients and their family members exhibit higher levels of corticosteroid phobia.^{30–33} However, given the complexity of the underlying factors, further research is warranted to investigate the reasons for these variations in corticosteroid use across countries and regions.

In Tianjin, the mean daily dosage of OCS among OCS users was 15.68 mg, exceeding the recommended dosage of 10 mg in Chinese guidelines.⁶ This suggests that some patients may endure the elevated risks of OCS side effects to manage their asthma, especially for patients with uncontrolled asthma, underscoring the need for better therapeutic options. Biologic therapies targeting immunoglobulin E and type 2 cytokines have been developed recently, with one of their benefits being the reduction of OCS use and even ICS use.³⁴ Although multiple biologics with different targets have been developed globally, only two—omalizumab and mepolizumab—were included in China's National Reimbursement Drug List, in 2020 and 2025, respectively.

Adherence to asthma control treatment in Tianjin was low, with only 17.78 % of patients exhibiting good compliance. The proportion is lower than that mostly derived from Europe, Australia, Canada, the United States, and Japan.^{24,28} Contributing factors may include subjective reasons like perceived symptom relief, and fear of side effects, alongside objective issues such as financial constraints and inconvenience of accessing medication, verified by another study in China.³⁵

The low adherence reveals a critical issue on severe asthma management in China. The mean age of patients with severe asthma was 56.15 years, a cohort potentially characterized by insufficient health education and age-related memory decline. In addition, there is a limit on the quantity of medication that can be dispensed at a single visit—commonly no more than a two-week supply, or up to one month under special circumstances—creating additional inconvenience for older adults. Furthermore, the substantial economic burden associated with severe asthma may compound the challenges of consistent adherence. Nevertheless, there is a paucity of comparative studies investigating adherence disparities between China and other countries, limiting a comprehensive understanding of the underlying causes.

The estimated all-cause total direct medical cost per patient in Tianjin was CNY 18,031.93, about 3.5 times the average total health expenditure per capita in China,³⁶ while in the US and Japan, it was approximately half and equal to their respective per capita expenditures.^{17,37–39} In addition, patients with uncontrolled asthma in

Tianjin incurred asthma-related total direct medical cost equivalent to 1.6 times the per capita health expenditure in China,³⁶ compared to only 40 % in the US.^{37,40} These findings reflect a greater economic burden for patients with severe asthma in Tianjin, especially for those with uncontrolled asthma.

More effective asthma management is needed to improve the condition. First, both patients and clinicians need better education on the importance of long-term asthma control and the use of corticosteroids.⁴¹ Second, loosening limits on prescription durations may enhance convenience and care continuity.⁴² Third, inclusion of severe asthma into outpatient chronic and special diseases coverage, a program in China providing higher reimbursement rates for covered disease—where severe asthma is not currently covered, could significantly alleviate patients' financial hardship.⁴³ Moreover, to improve access to more effective therapies, it is critical to facilitate the approval, reimbursement and broader clinical use of innovative biologic therapies.⁴⁴

Strengths and limitations

This study was conducted using a regional medical database with a large sample size in Tianjin, China. Unlike previous research, which has primarily focused on developed regions, our study provides novel insights from a developing country. This study indicates that, in China, patients with severe asthma have poorer adherence, a lower proportion of OCS usage but higher OCS doses among users, and experience a heavier economic burden. Furthermore, this study quantifies the economic burden of severe asthma across different asthma control levels, which is less investigated in previous studies, especially in developing countries. Overall, this study provides valuable quantitative insights to support decision-making by clinicians and policymakers. These findings are also informative for other developing countries with similar healthcare settings.

However, several limitations should be noted. First, a major limitation of this study is that asthma control was defined without direct measures such as symptom frequency, spirometric parameters, or validated composite scores—because these data were unavailable. Although the ATS/ERS guidelines recommend indirect definitions—an approach adopted in several existing studies—these methods are less sensitive than those based on direct measures. In addition, lack of detailed diagnostic process prevented confirming asthma cases (risking misclassification by clinicians with conditions like COPD) and distinguishing severe asthma from uncontrolled asthma due to poor adherence or incorrect inhaler technique. Second, the similar measures between asthma control and exacerbation inevitably introduce a correlation between them. Although GINA acknowledges that poor asthma

control is strongly linked to an increased risk of exacerbations, the association observed in our study should be interpreted with caution. Third, we assessed adherence using pharmacy-dispensing records, a method that often overestimates true adherence—a well-recognized limitation of database studies, especially for inhaled therapies, where the actual dose received hinges on correct inhalation technique. Fourth, because prescription drugs are also obtainable through community pharmacies in Tianjin, and such data are absent from our database, our adherence estimates could be biased. However, we believe the bias is relatively low in our study. The population we investigated consists of patients with severe asthma, who tend to obtain high-level consultation and care in hospitals, which is also confirmed by external clinical experts in Tianjin. Finally, the data used in this study were from 2018 to 2020. Since then, two biologics were added to the National Reimbursement Drug List in 2020 and 2025 respectively, which may influence the treatment options of severe asthma. However, the impact is likely to be minimal due to the currently low coverage of biologics.

Conclusions

Patients with severe asthma in Tianjin, China demonstrated poor medication adherence and a low proportion of OCS usage but higher OCS doses among users. These findings—combined with the observed tendency to seek alternative therapies and the substantial economic burden—underscore the urgent need to improve asthma medication management, particularly for patients with uncontrolled asthma.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Availability of data and material

All data generated or analyzed during this study are included in this published article and its supplementary information files. The additional datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

CRediT authorship contribution statement

Ke Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Xiaoning He:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Conceptualization. **Jing Wu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.hrtlng.2025.11.013](https://doi.org/10.1016/j.hrtlng.2025.11.013).

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